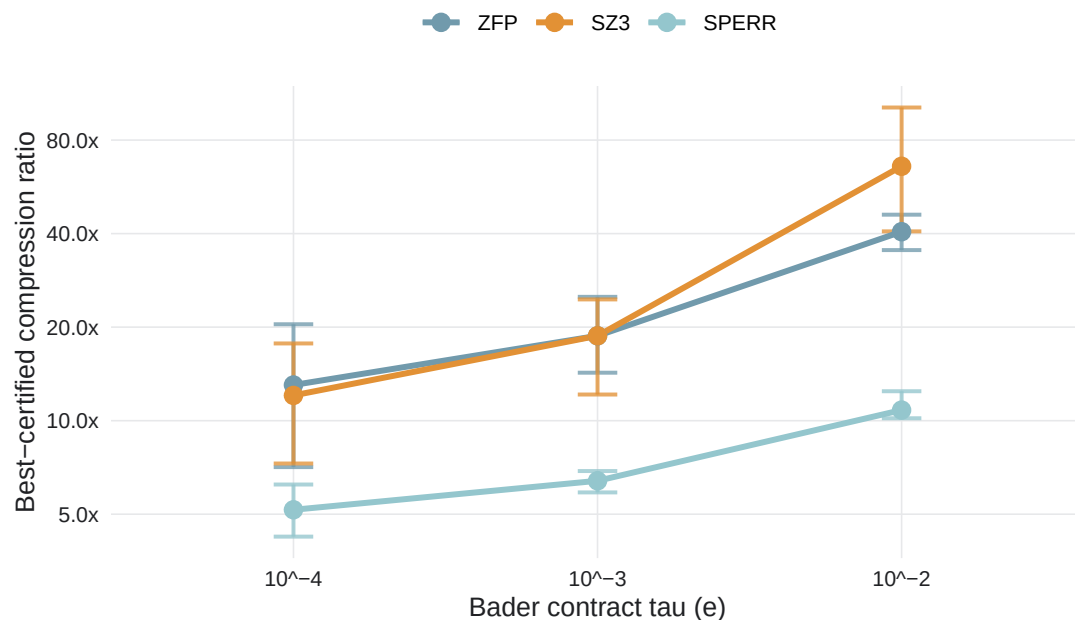


Figure 8 | Untouched external systems reproduce the stability-qualified decision frontier

A pre-frozen 63-system cohort preserves the codec-ordering transition, realized-distortion asymmetry, and QSQ eligibility logic without retuning.

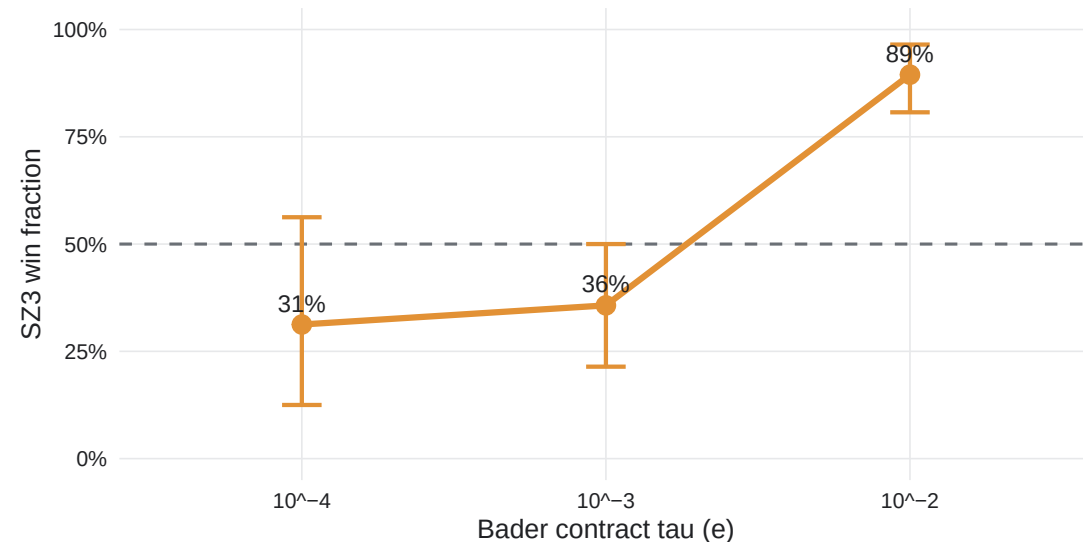
A | External systems reproduce the rate-fidelity transition

Median best-certified ratio; error bars are frozen bootstrap intervals



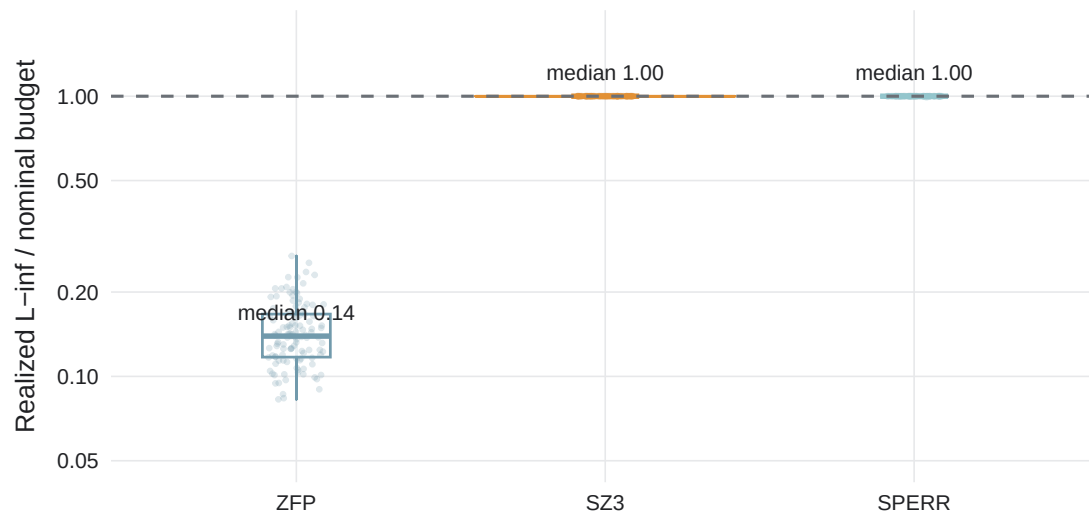
B | The ZFP-SZ3 ordering changes with scientific tolerance

Eligible external materials where SZ3 has the higher certified ratio



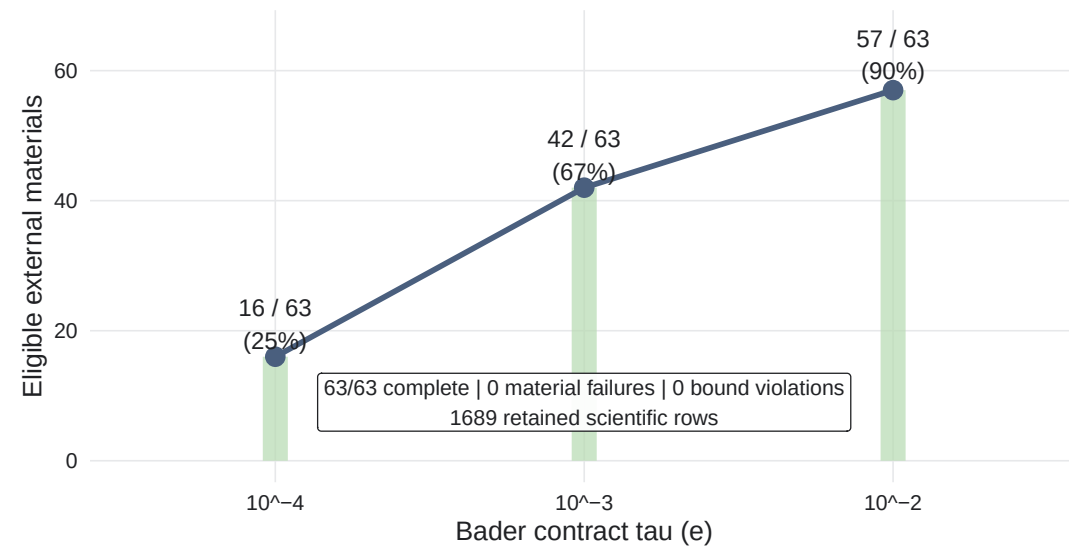
C | Nominal tolerance is not a common distortion scale

Certified external points; dashed line marks full use of the nominal budget



D | QSQ defines the measurable cohort

QSQ eligibility rises with requested Bader tolerance without retuning



External confirmatory cohort: 63/63 observed systems completed, 1,689 retained scientific rows, zero material-level pipeline failures and zero codec-bound violations.

Pairwise intervals and compression-ratio intervals are the frozen external bootstrap summaries.